UK NATIONAL RAIL PERFORMANCE, OPERATIONS & REVENUE INTELLIGENCE

National Rail

PROJECT HUB

PERFORMANCE & NETWORK

DELAY & OPERATIONS

REVENUE & CUSTOMER

ACTIONS

SEGMENTATION

DATA PERIOD

Jan 2024 – Apr 2024

PROJECT OVERVIEW

This project analyzes operational performance, passenger travel behavior, service reliability, and commercial performance across the UK National Rail network using journey data from January to April 2024. The dashboard integrates operational KPIs, delay intelligence, customer booking patterns, and revenue analytics to provide a comprehensive view of network performance and support data-driven operational and commercial decision-making.

The analytical frameworks were developed using histogram distribution analysis, percentile-based evaluation, and business rule validation to establish meaningful operational segments and ensure consistent performance reporting throughout the dashboard.

BUSINESS OBJECTIVES

- Evaluate overall railway network performance using operational, financial, and customer-focused KPIs.
- Identify delay patterns across routes, time periods, ticket classes, and operational causes to support service improvement initiatives.
- Analyze passenger booking behavior, ticket preferences, payment methods, and travel demand to understand customer purchasing patterns.
- Measure revenue performance across routes, booking types, railcards, payment methods, and time periods to identify high-performing business segments.
- Assess the relationship between operational performance, customer experience, and refund activity to understand the commercial impact of service disruptions.
- Standardize operational reporting through Time of Day, Booking Type, and Delay Severity frameworks for consistent business analysis.

DASHBOARD STRUCTURE

01

PERFORMANCE & NETWORK OVERVIEW

Overview of network performance, routes, revenue, trips, and on-time reliability across key dimensions.

02

DELAY & OPERATIONAL INTELLIGENCE

Deep dive into delays by time, severity, causes, and operational hotspots with incident analysis.

03

REVENUE & CUSTOMER INTELLIGENCE

Revenue performance, booking behavior, ticket preferences, and customer insights.

04

EXECUTIVE ACTION PLAN (TOP 10 RECOMMENDATIONS)

Top 10 strategic recommendations to improve performance, reduce delays, and drive revenue growth.

05

SEGMENTATION FRAMEWORK & BUSINESS DEFINITIONS

Detailed frameworks, segmentation logic, and business definitions used in the analysis.

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Performance & Network Overview

Month: All

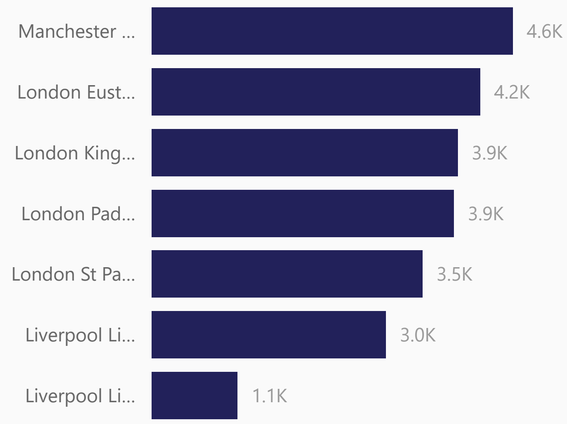
Ticket Class: All

Time of Day: All

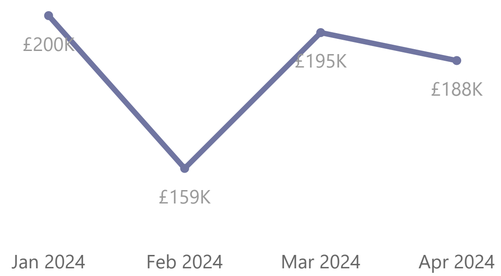
Route: All

Total Routes	Total Revenue (£)	Total Trips	Revenue Per Trip	On-Time Rate	Refund Rate
65	£741.9K	32K	23.4	93%	3.5%

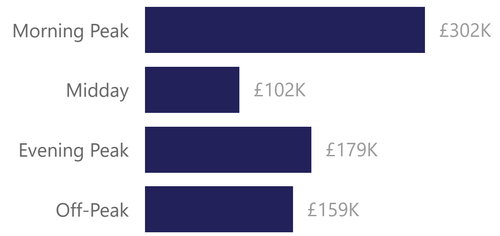
Top Routes by Trip Volume



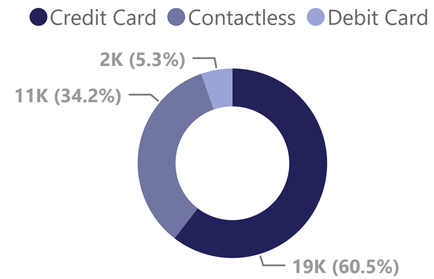
Revenue Trend



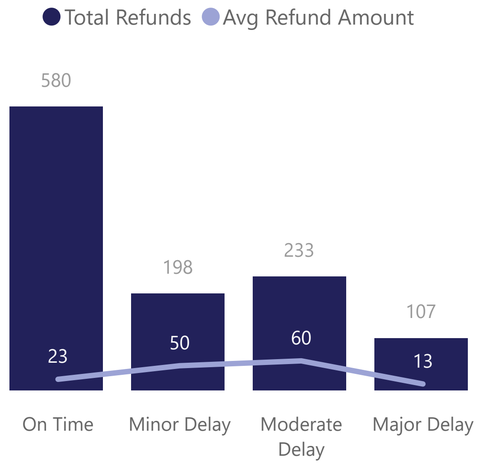
Revenue by Time of Day



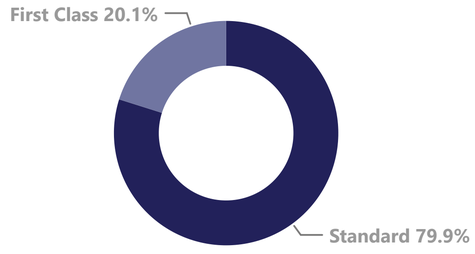
Trip Distribution by Payment Method



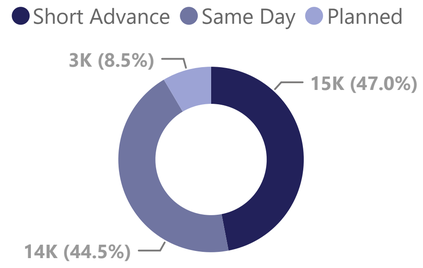
Refund Trends Across Delay Severity



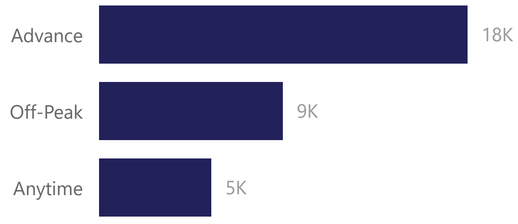
Revenue by Ticket Class



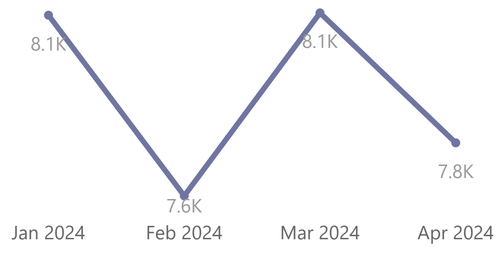
Trip Distribution by Booking Type



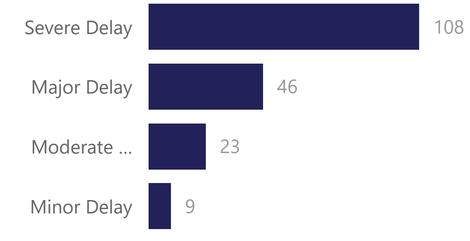
Trip Distribution by Ticket Type



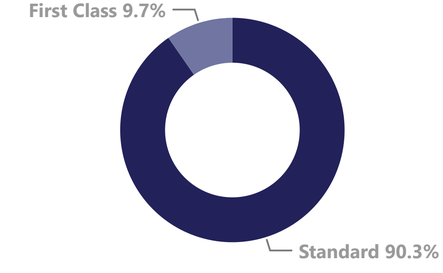
Trip Volume Trend



Delay Duration by Severity



Trip Distribution by Ticket Class



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Delay & Operational Intelligence

Month: All

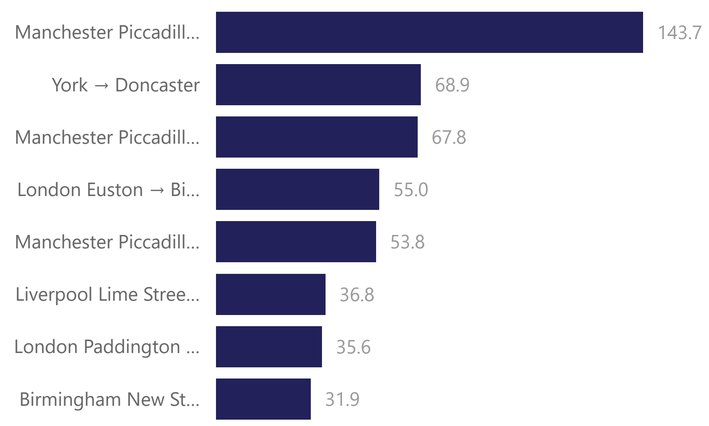
Ticket Class: All

Time of Day: All

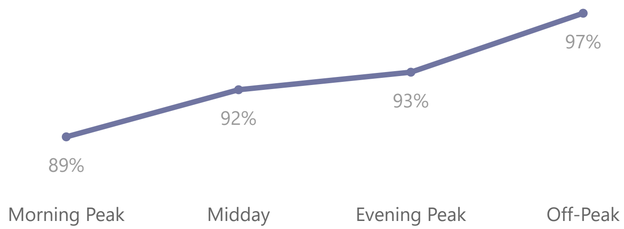
Route: All

Average Delay (Minutes)	Critical Delay Threshold	Above Average Delay (%)	Delay Rate	Severe Delay Share (%)
42.5	131	42%	7.2%	14%

Top Routes by Average Delay

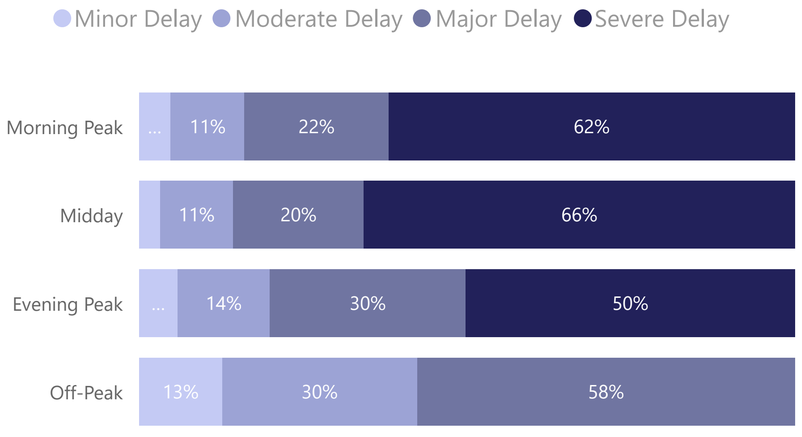


On-Time Performance by Time of Day



Severe Delay Share by Ticket Class

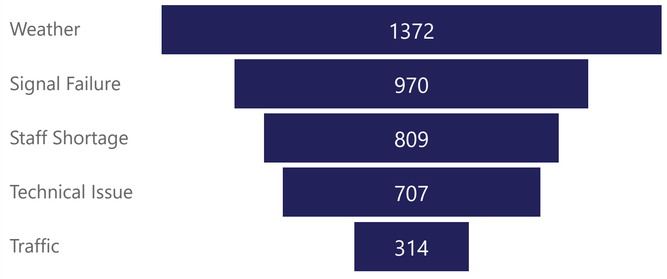
Delay Distribution by Time of Day



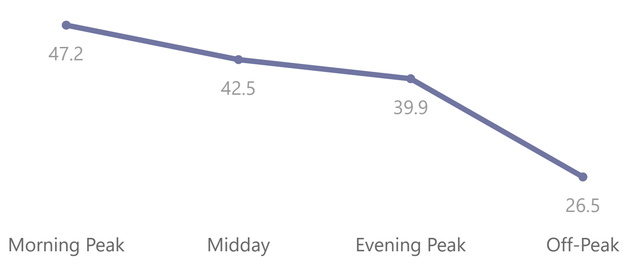
Delay Hotspots by Time Bucket

Delay Reason	Morning Peak	Midday	Evening Peak	Off-Peak
Signal Failure	85	19	28	30
Staff Shortage	40	59	50	31
Technical Issue	32	27	21	
Traffic	21		54	
Weather	41		74	23

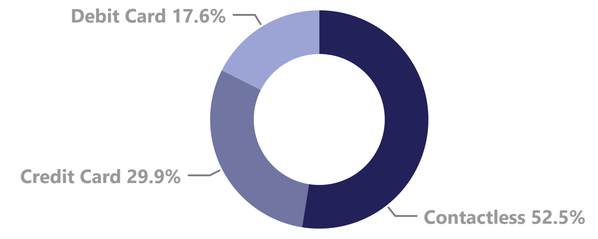
Delay Incidents by Reason



Average Delay by Time of Day



Delay Share by Payment Method



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Revenue & Customer Intelligence

Month: All

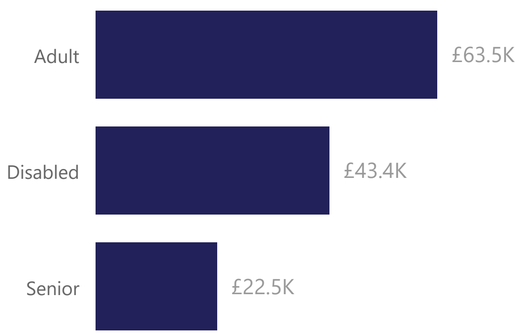
Ticket Class: All

Time of Day: All

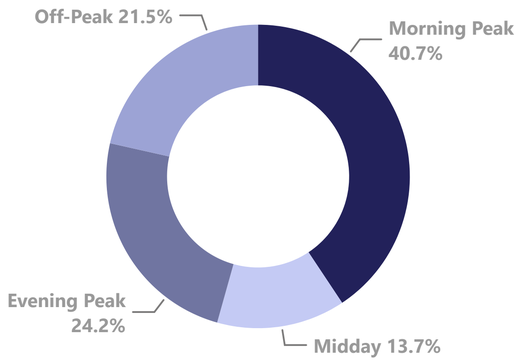
Route: All

Total Revenue (£)	Total Trips	Revenue per Trip (£)	Total Refunds	Top Revenue Route
£741.9K	32K	23.4	1.12K	London Kings Cross → York

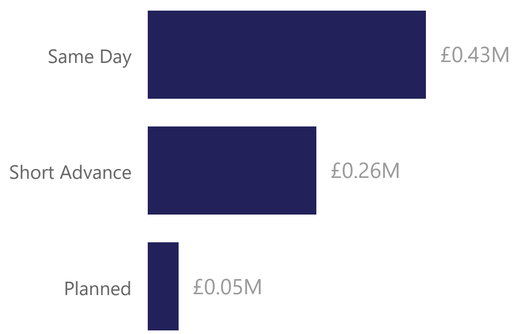
Revenue by Railcard Type



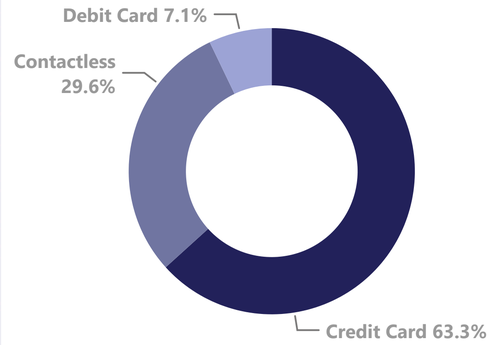
Revenue Distribution by Time of Day



Revenue by Booking Type



Revenue Distribution by Payment Method



Top 10 Route Performance Summary

Route	Revenue (£)	Trips	Revenue per Trip	On-Time %	Avg Delay (minutes)	Refunds
London Kings Cross → York	£183K	3922	47	97%	16	82.00
Liverpool Lime Street → London Euston	£113K	1097	103	29%	37	171.00
London Paddington → Reading	£65K	3873	17	98%	36	89.00
London Euston → Manchester Piccadilly	£61K	712	86	100%		21.00
London St Pancras → Birmingham New Street	£53K	3471	15	100%		64.00
London Euston → Birmingham New Street	£50K	4209	12	94%	55	83.00
Manchester Piccadilly → London Euston	£34K	345	98	32%	25	96.00
Birmingham New Street → London St Pancras	£19K	702	27	100%		12.00
Manchester Piccadilly → Liverpool Lime Street	£17K	4628	4	92%	68	91.00
Manchester Piccadilly → London Paddington	£16K	144	114	100%		3.00

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EXECUTIVE ACTION PLAN; TOP 10 RECOMMENDATIONS

OUR STRATEGIC PRIORITY

Maximize sustainable revenue and customer trust by focusing on high-impact actions that enhance reliability, reduce disruption, and elevate customer experience.

01 Protect Morning Peak Revenue

Prioritize Morning Peak reliability—it generates 40.7% of network revenue (£301.65K) but has the lowest punctuality (89%) and highest delay rate (11.07%).

KPI Revenue Share 40.7% | On-Time 89%

02 Turn Around the Liverpool–London Corridor

Prioritize Liverpool Lime Street → London Euston, where high yield and recurring revenue are exposed to just 20%–36% monthly punctuality.

KPI Rev/Trip £91–£109 | On-Time 20–36%

03 Recover Manchester–London High-Yield Value

Address chronic unreliability on Manchester Piccadilly → London Euston, where revenue reaches £153 per First Class trip despite only 22% punctuality.

KPI Rev/Trip (1st Class) £153 | On-Time 22%

04 Reduce Midday Refund Leakage

Target Midday service failures—it generates the lowest revenue (£101.59K) yet produces the highest refund rate (7.03%) and 407 refunds.

KPI Refund Rate 7.03% | Refunds 407

05 Match Delay Actions to Root Causes

Deploy targeted interventions by operating window: strengthen weather resilience in Morning Peak, staffing in Midday, and signal reliability in Evening Peak.

KPI Top Causes by Time of Day

06 Close the First Class Reliability Gap

Protect premium value: First Class generates 2.36x Standard revenue per trip, yet records worse performance across all key reliability metrics.

KPI Rev/Trip £48.86 vs £20.72 | On-Time 92% vs 93%

07 Prevent Delays from Becoming Severe

Manage delay frequency and escalation separately—Morning Peak has highest delay rate, while Evening Peak carries the highest severe delay share (25%).

KPI Severe Delay Share 25% (Evening Peak)

08 Improve Manchester–Liverpool Volume Corridor

Prioritize service resilience on Manchester Piccadilly → Liverpool Lime Street, a high-volume route with only ~£4 revenue per trip and 61–74 min monthly delays.

KPI Trips ~1.1K–1.2K | Rev/Trip ~£4 | Delay 61-74 min

09 Replicate Kings Cross–York Success Model

Use London Kings Cross → York as the internal benchmark, combining revenue leadership with 95%–100% punctuality across time periods.

KPI On-Time 95%–100% | Revenue Leader

10 Diagnose & Reverse February's Lost Yield

Investigate February's 15.3% revenue-per-trip decline, as revenue fell 20.2% while trips declined only ~6%, pointing to a commercial mix issue.

KPI Rev -20.2% | Trips -6% | Rev/Trip -15.3%

EXPECTED OUTCOMES

- Stronger Revenue Protection
- Higher Customer Satisfaction
- Lower Delay & Refund Losses
- Smarter Resource Allocation
- Sustainable Profitable Growth

SEGMENTATION FRAMEWORK & BUSINESS DEFINITIONS

METHODOLOGY

The analytical frameworks were developed through a combination of **histogram distribution analysis, percentile-based evaluation**, and **business rule validation** to identify natural segmentation thresholds within the railway dataset. Rather than using arbitrary cut-offs, each framework was designed around observed data distributions and operational characteristics, ensuring meaningful classification for demand analysis, booking behavior assessment, delay severity evaluation, and service performance monitoring.

TIME BUCKET FRAMEWORK

TIME BUCKET

Segment	Journey Hour
Morning Peak	06:00 – 09:59
Midday	10:00 – 19:59
Evening Peak	16:00 – 19:59
Off-Peak	Remaining Hours

ANALYTICAL APPLICATIONS

- Standardizes railway operations into meaningful service periods.
- Enables demand and revenue comparison across operational windows.
- Supports peak hour performance and delay analysis.
- Helps identify service patterns throughout the day.

DELAY SEVERITY FRAMEWORK

DELAY STATUS

Segment	Delay Duration
On Time	0 Minutes
Minor Delay	1 – 15 Minutes
Moderate Delay	16 – 60 Minutes
Major Delay	31 – 60 Minutes
Severe Delay	More than 60 Minutes

ANALYTICAL APPLICATIONS

- Standardizes delay categorization by operational severity levels.
- Supports delay trend and performance monitoring.
- Enables comparison of service quality across routes and time periods.
- Helps identify operational risks and prioritize service improvement initiatives.

BOOKING TYPE FRAMEWORK

BOOKING TYPE

Segment	Booking Days
Same Day	0 Days
Short Advance	1 – 5 Days
Planned	More than 5 Days

ANALYTICAL APPLICATIONS

- Standardizes customer booking behavior into meaningful booking segments.
- Supports demand forecasting based on passenger planning habits.
- Enables revenue comparison across booking categories.
- Helps evaluate purchasing behavior and travel planning trends.

WHY THESE FRAMEWORKS MATTER

These segmentation frameworks form the foundation of our end-to-end analysis, enabling consistent classification, meaningful insights, and data-driven decision-making across revenue, operations, customer behavior, and network performance.